

Thermal-Electrical-Mechanical Performance Enhancement in TSV: A Multi-objective Optimization Approach

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Abstract—In this study, the multi-physics simulations of Through Silicon Via (TSV) are carried out based on HFSS and Icepak, and then corresponding surrogate model is established based on deep neural network (DNN). On this basis, multi-objective optimization design was carried out based on Non-dominated Sorting Genetic Algorithm III (NSGA-III algorithm), and the Pareto solution of the design parameters were obtained. The findings demonstrate that, relative to the initial design scheme, the optimized solutions can enhance insertion loss by as much as 60.03%, cut down the temperature by as much as 9.09%, and lessen the thermal strain by as much as 33.18%. Additionally, when considering the 16 optimized solutions collectively, they present mean enhancement percentages of 28.52%, 9.28%, and 29.31% with regard to insertion loss improvement, temperature reduction, and thermal strain reduction, respectively. These outcomes confirm that the multi-objective optimization approach has successfully balanced the competing demands of TSV signal integrity, thermal management, and mechanical reliability design, which can provide certain references for related studies.

Keywords—Through Silicon Via (TSV); Multi-objective optimization; Non-dominated Sorting Genetic Algorithm III (NSGA-III algorithm); Deep neural network (DNN); Multi-physics simulations

I. INTRODUCTION

In the realm of semiconductor innovation [1], the relentless pursuit of enhanced performance, reduced power consumption, and miniaturization has driven the evolution of chip design [2]. As planar architectures reach their physical limits, the transition to three-dimensional (3D) architectures has become imperative [3]. This shift introduces not only the promise of higher integration density and shorter interconnects but also the complexity of managing thermal and electrical effects in compact spaces [4]. Through Silicon Via (TSV) have emerged as a critical enabler for 3D integration [5]. These vertical interconnects facilitate the

stacking of multiple silicon layers, reducing chip footprint and enabling faster signal transmission. However, the intricate nature of 3D designs, coupled with the need to address thermal and mechanical reliability, presents a multifaceted challenge that requires innovative solutions [6]. Currently, reliability of TSV has become a focal point of research. TSV operate in increasingly demanding environments, necessitating solutions that can withstand thermal fluctuations, electrical stresses, and mechanical vibrations [7]. Prior work has investigated thermal management strategies to mitigate heat dissipation issues [8] and explored novel materials to enhance mechanical resilience [9]. Signal integrity has been scrutinized to ensure reliable communication across densely packed interconnects [10]. Despite these efforts, the true complexity of TSV lies in their multi-physics nature, where thermal, electrical, and mechanical factors interplay. This complexity demands a holistic approach that transcends single-parameter optimization.

Multi-objective optimization offers a systematic approach to navigate this complexity [11]. It allows designers to balance multiple performance indicators simultaneously. In the realm of TSV multi-objective optimization design, the process may entail optimizing the TSV diameter to reduce resistance. At the same time, careful consideration must be given to issues such as the mismatch of thermal expansion between different materials. The ultimate goal is to achieve an optimal design that best balances these competing objectives. Recent studies have demonstrated the potential of the multi-objective optimization method. For example, Huang et al. [12] successfully developed an optimization framework that seamlessly integrates finite element analysis with advanced algorithms like NSGA-II. This powerful combination enables effective exploration of the design space, and provides a set of designs that represent the best possible compromises among the various objectives. Wang et al. [13]

derived the regression equations for thermodynamic coupling stress, random vibration stress, and return loss based on the response surface method, and then employed the entropy weight theory to calculate the weight coefficients and ultimately obtained the optimal parameter combination. Beyond thermodynamic coupling issues, the high-frequency signal integrity of TSV has also become a research hotspot. Liu et al. [14] proposed a novel collaborative optimization framework based on the multi-output multi-fidelity surrogate model (MOMFSM), which has been proven to be highly effective in achieving multi-objective optimization design of TSV. Experimental results indicate that it can significantly enhance prediction accuracy at a relatively low computational cost. He et al. [15] constructed the combined fitting objective function for the experimental group through response surface analysis, and then applied the Technique for Order Preference by Similarity to an Ideal Solution (TOPSIS) to assign weights to the two optimization objectives, ultimately realize the 13.8% reduction of thermal resistance and 9.74% improvement of signal transmission efficiency. To address the heat dissipation challenges posed by the high-power density, Xu et al. [16] carried out the study on the performance optimization of the microneedle finned channel cooling system embedded in coaxial TSV, and investigated the influences of TSV aspect ratio and pitch ratio on shielding efficiency and heat transfer performance. Results reveal that when the TSV length-to-diameter ratio is 15 and the spacing ratio is 2.5, the maximum temperature decreases by 3.97%. These studies provide a good reference for the multi-objective optimization design of TSV.

However, there is still room for advancement. Previous studies have often relied on simplified models that do not fully capture the multi-physics interactions within TSV. Based on these issues, in this study, the multi-physics simulations of TSV are carried out, and corresponding surrogate model is established based on deep neural network (DNN). On this basis, multi-objective optimization was carried out based on Non-dominated Sorting Genetic Algorithm III (NSGA-III algorithm), and the Pareto solution of the design parameters were obtained. Results show that multi-objective optimization method can well simultaneously improve and balance the competitive requirements of TSV signal integrity, thermal management and mechanical reliability design, and can provide certain references for related research.

II. RELEVANT THEORIES

The spatial potential distribution among TSV can be defined as $\phi(\mathbf{x})$, and current density $\mathbf{J}$ can be expressed as:

$$\mathbf{J} = -\sigma(\mathbf{x})\nabla\phi \quad (1)$$

where $\sigma(\mathbf{x})$ is the conductivity at position $\mathbf{x}$, so the local Joule heat generation power can be expressed as followed.

$$q_e = \sigma(\mathbf{x})(\nabla\phi)^2 \quad (2)$$

The thermal equilibrium among TSV is determined by the heat conduction equation.

$$\nabla \cdot (\kappa \nabla T) + q_e = \rho c \frac{\partial T}{\partial t} \quad (3)$$

where ∇ represents the gradient operator, κ , q_e , ρ and c denote the thermal conductivity, heat flux, material density and specific heat capacity, respectively. Thermal expansion result from temperature change can be calculated based on followed equation.

$$\varepsilon^{th} = \alpha \Delta T \mathbf{I} \quad (4)$$

where α and ΔT represent thermal expansion coefficient and temperature increment, and $\mathbf{I}$ denotes the second-order unit tensor.

During the signal transmission, resistance, inductance, conductance, and capacitance of TSV can be calculated as followed.

$$R = 1 / \sigma A \quad (5)$$

$$L = \mu P / A \quad (6)$$

$$G = A \omega \epsilon'' / d \quad (7)$$

$$C = A \epsilon' / d \quad (8)$$

Based on the above derivation, the characteristic impedance evolution Z_0 can be obtained.

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}} \quad (9)$$

where j denote the imaginary unit. Accordingly, insertion loss IL can be obtained.

$$IL = 20 \log_{10} \left| \frac{2Z_0}{Z_0 + Z_L} e^{-\gamma l} \right| \quad (10)$$

where Z_L and Z_S represents the load impedance and source impedance, respectively.

III. MULTI-OBJECTIVE OPTIMIZATION BASED ON SURROGATE MODEL

A. Construction of TSV model

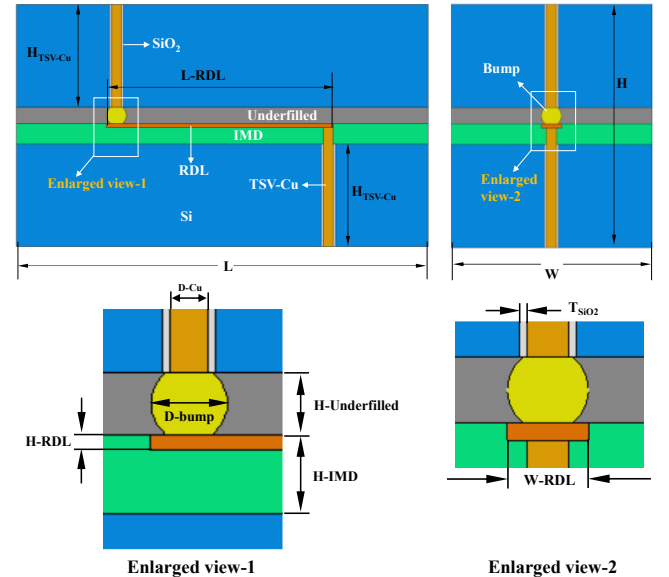


Fig 1. Description of the structure and dimensional parameters of TSV.

This study focuses on the interconnected TSV in a three-dimensional chip. As depicted in Fig 1, TSV is mainly consists of Cu column, SiO₂ insulation layer, silicon substrate, bump, redistribution layer (RDL), intermetallic dielectric layer (IMD) and underfilled layer. The Cu column is fabricated via electroplating process, the RDL can be

obtained through deposition process. IMD and underfilled layers adopt polyimide and epoxy, respectively. All material parameters are listed in TABLE I. The initial design scheme of TSV is derived from study [17], which have been acquired the optimal design case of TSV based on interface fatigue analysis. On the basis, multi-objective optimization design will be carried out subsequently to seek the optimal solution of trade-offs among various performances. The design parameters can be divided into three categories, called

independent parameters, dependent parameters and fixed parameters. During the optimization process, the independent parameters can seek the optimized solution within the set interval, dependent parameters are subordinate to independent parameters, and fixed parameters remain unchanged. Therefore, five independent parameters are selected as the optimization variables, and corresponding optimization interval are assigned as shown in TABLE II.

TABLE I. MAIN MATERIAL PARAMETERS

Parameters	Density	Young's modulus	Poisson's ratio	Coefficient of thermal expansion	Specific heat	Thermal conductivity	Electric conductivity	Dielectric constant
unit	kg/m ³	Pa	/	K ⁻¹	J/(kg. K)	W/(m.K)	S/m	/
Si	2300	1.3E11	0.28	3.00E-6	712	157	2.5E-04	11.9
SiO ₂	2270	7.0E10	0.16	6.00E-7	745	1.4	1.0E-14	4
TSV-Cu	8900	7.0E10	0.34	1.82E-5	385	398	5.6E+7	/
RDL	8900	1.1E11	0.35	1.72E-5	385	398	5.6E+7	/
Bump	8960	1.3E11	0.34	1.70E-5	385	385	5.8E+7	/
IMD	1400	3.8E9	0.3	3.20E-5	1510	0.19	1.0E-15	3.2
Underfilled layer	1200	4.1E9	0.3	3.50E-5	1600	0.5	1.0E-15	3.5

TABLE II. INITIAL SCHEME AND OPTIMIZATION PARAMETER SETTING

Parameter	Implication	Parameters type	Initial Value	Optimization interval	Final Value Expression
D_{Cu}	Diameter of Cu column	Independent	5μm	[4,6]	D_{Cu}
H_{TSV-Cu}	Height of TSV-Cu	Independent	50μm	[45,55]	H_{TSV-Cu}
T_{SiO2}	Thickness of SiO ₂ layer	Independent	1μm	[0.8,1.2]	T_{SiO2}
D_{Bump}	Diameter of Bump	Independent	10μm	[9,11]	D_{Bump}
P	Pitch between TSVs	Fixed	100μm	/	100μm
H_{Underfilled}	Height of Underfilled layer	Dependent	8μm	/	0.8* D_{Bump}
L_{RDL}	Length of RDL layer	Dependent	107.5	/	P +0.5*(D_{Bump} + D_{Cu})
W_{RDL}	Width of RDL layer	Dependent	10μm	/	D_{Bump}
H_{RDL}	Height of RDL layer	Fixed	2μm	/	2μm
H_{IMD}	Height of IMD layer	Fixed	10μm	/	10μm
L	Length of Si substate	Dependent	200μm	/	P +100
W	Width of Si substate	Dependent	100μm	/	P
H	Height of Si substate	Dependent	118μm	/	H_{IMD} + H_{Underfilled} +2* H_{TSV-Cu}

B. Surrogate model construction

Multi-objective optimization demands extensive test or simulation data, which is computationally intensive and time-consuming. To address this problem, surrogate model construction offers a cost-efficient alternative. The construction process of the surrogate model is illustrated in Fig 2 and described as followed.

- **Simulation:** The design parameters of TSV are sampled in the design space based on Latin Hypercube Sampling method (LHS), and 500 representative data points are collected. Based on the sampled design parameters, simulations are carried out based on HFSS and Icepak. In terms of electromagnetic calculation, the sweep frequency range is set to 0-50GHz, and two TSV-Cu are assigned as wave port attribute. In terms of thermal-

mechanical calculation, the initial temperature of the solution domain is set at 20°C, and one of the surfaces perpendicular to the ground is taken as the air flow inlet, with the temperature and flow velocity of 50°C and 0.1m/s respectively. The remaining four surfaces are defined as walls without heat exchange.

- **Surrogate Model Training:** Standardize the simulation results (such as insertion loss, temperature, and thermal strain) and split them into a training dataset and a validation dataset in a 4:1 ratio. Construct a surrogate model using a DNN with an input layer sized of 13 (correspond to the number of design parameters), two hidden layers comprising 64 and 32 neurons with ReLU activation, respectively, and an output layer with 3 linear neurons for regression. Adam optimizer is selected for model training with a

learning rate of 0.001, and the batch size and the maximum number of epochs are set as 32 and 200, respectively.

- **Model Validation:** The accuracy of the surrogate model is verified based on the verification dataset. The specific equation is as followed.

$$R^2 = \left(1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y}_i)^2} \right) \quad (11)$$

where y_i , $\hat{y}_i$ and $\bar{y}_i$ represents the simulation value, the mean value of and the prediction value of insertion loss, temperature and thermal strain. Results show that the prediction accuracies of the surrogate model for insertion loss, temperature and thermal strain are 89.22%, 92.17% and 90.25%, respectively, indicating that the surrogate model has sufficient accuracy and can be used for the subsequent multi-objective optimization.

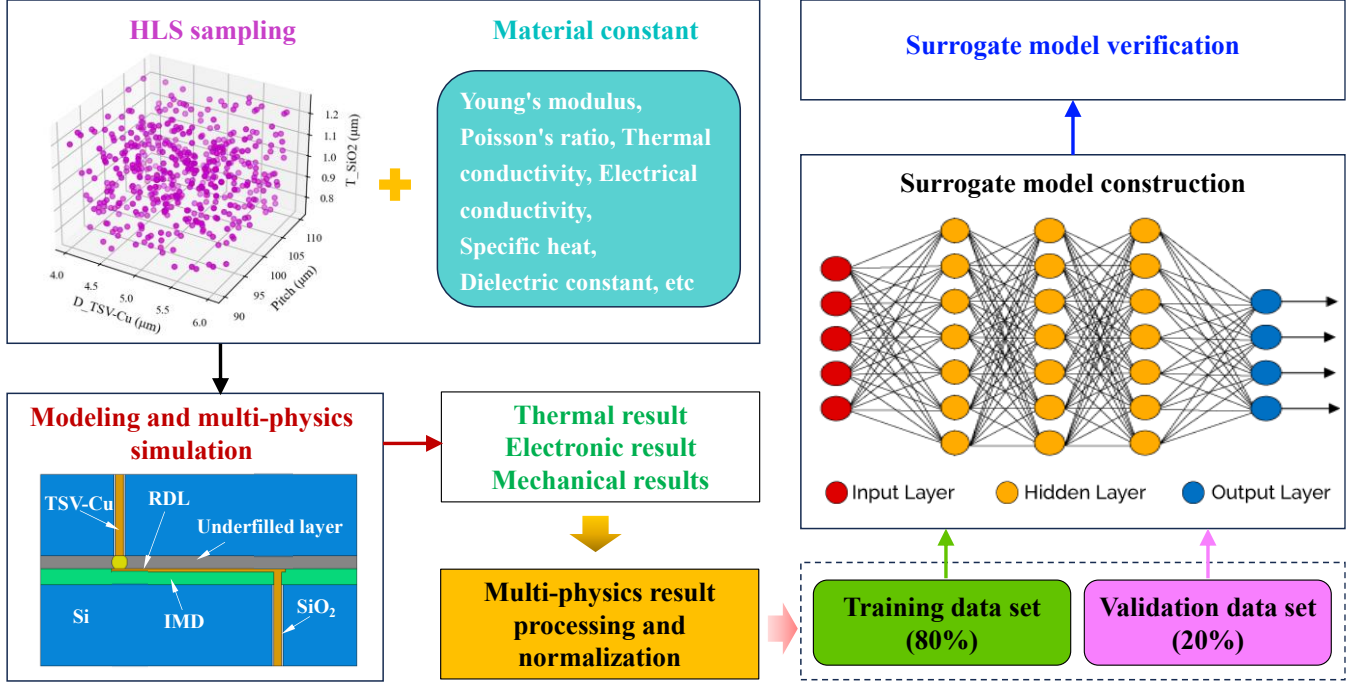


Fig 2. Multi-objective optimization technical route of TSV.

C. Multi-objective optimization based on NSGA-III

Based on the created surrogated model, multi-optimization can be carried. The objective of optimization is to maximize the insertion loss and minimize the temperature rise and thermal strain of TSV, corresponding optimization problem can be expressed as followed.

$$\begin{aligned} \text{Minimize } f(\mathbf{x}) &= [-f_1(\mathbf{x}), f_2(\mathbf{x}), f_3(\mathbf{x})]^T \\ \text{Subject to } \mathbf{x} &\in \Omega \end{aligned} \quad (12)$$

where $\mathbf{x}$ represents the design parameters, $f_1(\mathbf{x})$, $f_2(\mathbf{x})$ and $f_3(\mathbf{x})$ represent the objective functions of insertion loss, temperature and thermal strain of TSV, respectively. Ω is the feasible design space, which can be obtained from TABLE II.

Multi-objective optimization can be performed using the NSGA-III algorithm. It ranks solutions via non-dominated sorting based on their non-domination level and calculates crowding distance to preserve population diversity. Parents are selected according to rank and crowding distance, after which crossover and mutation generate offspring. The NSGA-III selection operator seeks solutions near reference points while maintaining diversity by minimizing the perpendicular distance from solutions to the reference lines. The perpendicular distance from a solution $f(\mathbf{x})$ to a reference line is calculated as followed.

$$d(f(\mathbf{x}), \mathbf{r}_i) = \frac{\|f(\mathbf{x}) - \mathbf{r}_i\|}{\|\mathbf{w}_i\|} \quad (13)$$

where $\mathbf{r}_i$ is the i^{th} reference point and $\mathbf{w}_i$ is the corresponding weight vector.

IV. RESULTS

After multiple iterations, a set of Pareto optimal solutions was obtained, representing the best trade-offs among conflicting objectives. Fig 3 presents 16 Pareto solutions and their corresponding main performance. Compared to the initial design scheme, all optimized solutions show performance improvements. For instance, the red-marked optimized solution has an insertion loss of -0.25 dB, a maximum temperature of 22.00 °C, and a maximum thermal strain of 3.13e-5, all better than the initial values of -0.40 dB, 24.08 °C, and 4.44e-5.

Table III lists the independent design parameters corresponding to the Pareto solution set. Table IV records the performance improvement of the Pareto solution set over the initial solution in terms of insertion loss, temperature reduction, and thermal strain reduction. The results show that the Pareto solution set can greatly enhance the overall performance of TSV. For example, solution 9 has the largest insertion loss improvement of 60.03%, solution 16 has the largest temperature reduction of 9.09% and thermal strain reduction of 33.18%.

Moreover, the 16 optimized solutions have average improvement degrees of 28.52%, 9.28%, and 29.31% in insertion loss, temperature reduction, and thermal strain reduction, respectively, which indicates that the multi-objective optimization process has effectively achieved the competing goals of TSV signal integrity, thermal management, and mechanical reliability design.

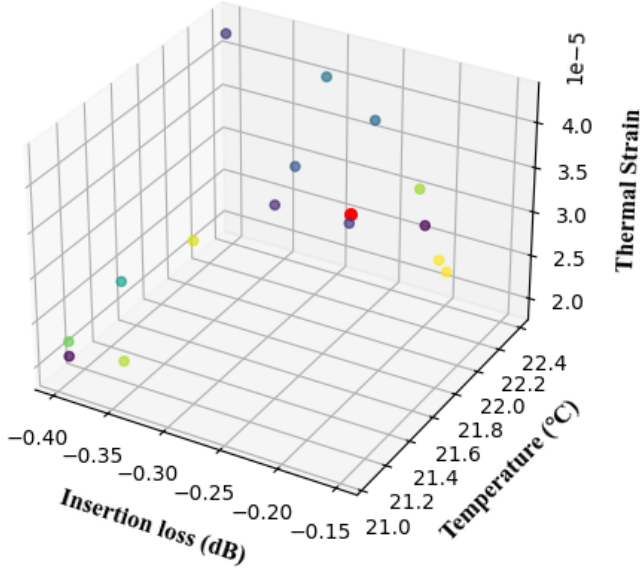


Fig 3. Pareto solution set.

TABLE III. PARETO SOLUTIONS OF INDEPENDENT DESIGN PARAMETERS

Solution ID	D_{Cu}	H_{TSV-Cu}	T_{SiO_2}	D_{Bump}
1	4.78	54.17	1.13	9.07
2	5.87	53.62	0.97	10.32
3	5.87	50.85	0.97	10.87
4	5.81	48.15	0.92	10.71
5	5.85	48.22	0.87	10.55
6	4.78	54.16	1.13	9.07
7	4.84	50.45	1.08	9.59
8	5.24	53.55	0.97	10.29
9	5.91	50.00	0.96	10.92
10	5.80	50.10	0.95	10.62
11	5.91	49.91	0.96	10.82
12	4.95	51.89	1.10	9.12
13	5.87	52.43	1.00	10.06
14	5.11	50.46	1.10	9.20
15	5.90	49.98	0.97	10.81
16	5.92	49.91	0.96	10.92

TABLE IV. PERFORMANCE IMPROVEMENT OF THE PARETO SOLUTION COMPARED TO THE INITIAL SCHEME

Solution ID	Insertion loss improvement	Temperature reduction	Thermal strain reduction
1	0.79%	12.49%	52.83%
2	27.01%	9.49%	25.31%
3	43.03%	9.42%	25.24%

4	33.69%	7.00%	16.94%
5	25.05%	8.15%	10.06%
6	5.09%	10.02%	40.06%
7	7.91%	11.76%	56.72%
8	10.00%	9.74%	38.34%
9	60.03%	8.99%	31.72%
10	38.21%	8.59%	29.60%
11	56.17%	8.87%	25.03%
12	2.04%	6.14%	5.15%
13	24.83%	8.46%	22.64%
14	6.60%	11.59%	37.84%
15	53.48%	8.63%	18.31%
16	62.35%	9.09%	33.18%

V. CONCLUSIONS

In this study, multi-objective optimization of TSV was carried out based on DNN and NSGA-III algorithm, achieving the improvement of the comprehensive performance of TSV. Based on the above analysis, the following conclusions can be drawn:

1) Multi-physics simulations of TSV are carried out based on HFSS and Icepak, and corresponding surrogate model is established based on DNN.

2) Multi-objective optimization is carried out based on NSGA-III algorithm, and the Pareto solution of design parameters were obtained.

3) The optimized solutions can improve insertion loss by up to 60.03%, reduce temperature by up to 9.09%, and decrease thermal strain by up to 33.18%.

4) The 16 optimized solutions present mean enhancement percentages of 28.52%, 9.28%, and 29.31% with regard to insertion loss improvement, temperature reduction, and thermal strain reduction, respectively.

In the future, research in multi-objective optimization of TSV will focus on improving optimization algorithms and balancing multiple conflicting objectives.

ACKNOWLEDGMENT

This research work is sponsored by the National Natural Science Foundation of China (No. 52275166), Youth Independent Innovation Science Fund of NUDT (ZK23-29), Fund for State Key Laboratory (6142003202309; 6142806230303) and Hunan Graduate Research Innovation Project (CX20230020).

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